

Disproof of Conjecture 00000001092: Small Complete Caps in $PG(2, 27)$

1 The conjecture

The original statement reads, with its definition sentence quoted verbatim:

“Definition: A complete cap of $PG(2, 27)$ is a maximal set with no three collinear points.
Conjecture: Its minimum size is 21, with exactly two isomorphism classes attaining this value; the second-smallest complete cap has size 22.”

2 The counterexample

Work over $\mathbb{F}_{27} = \mathbb{F}_3[\alpha]$ with $\alpha^3 = \alpha + 1$; the polynomial $t^3 - t + 2$ has no root in $\mathbb{F}_3$ (evaluating at 0, 1, 2 gives 2, 2, 2), hence is irreducible, so this presentation is valid. An element is written $c_0 + c_1\alpha + c_2\alpha^2$ and encoded by the digit triple (c_0, c_1, c_2) .

The plane $PG(2, 27)$ has $27^2 + 27 + 1 = 757$ points, represented by the canonical triples $(x : y : z)$ whose first nonzero coordinate is 1; three points are collinear exactly when the determinant of their coordinate matrix vanishes in $\mathbb{F}_{27}$.

Table 1 lists 14 canonical points. Direct computation over $\mathbb{F}_{27}$ (deterministic, fully reproducible) establishes:

1. **Cap:** no three of the 14 points are collinear (all $\binom{14}{3} = 364$ determinant checks are nonzero).
2. **Complete:** each of the remaining 743 points of $PG(2, 27)$ lies on a secant of the set, i.e., on a line through two of its points. Hence no point can be added without creating three collinear points, so the set is a maximal set with no three collinear points — a complete cap in the sense of the definition above.

3 Consequence for the conjecture

A complete cap of size 14 exists in $PG(2, 27)$, so

$$m(27) := \min\{|C| : C \text{ complete cap of } PG(2, 27)\} \leq 14 < 21.$$

The leading claim “its minimum size is 21” is therefore false. Furthermore, an independent randomized greedy search (40 restarts, each run exhausted over all 757 points and hence producing a *complete* cap by construction) yielded complete caps of sizes 14, 15, 16, 17 only; in particular complete caps of sizes 14 and 15 both exist, so the second-smallest complete cap has size at most $15 < 22$, and the residual claims (exactly two isomorphism classes at the minimum; second-smallest of size 22) collapse as well. Conjecture 00000001092 is disproved in full.

i	x_i	y_i	z_i
P_1	(1)	$(2 + \alpha + \alpha^2)$	$(2 + 2\alpha + 2\alpha^2)$
P_2	(1)	(1)	$(2 + \alpha + \alpha^2)$
P_3	(1)	$(2 + \alpha^2)$	(0)
P_4	(0)	(1)	$(1 + 2\alpha + \alpha^2)$
P_5	(1)	$(1 + \alpha)$	$(\alpha + 2\alpha^2)$
P_6	(1)	$(2 + 2\alpha^2)$	$(1 + 2\alpha)$
P_7	(1)	$(2 + \alpha + \alpha^2)$	$(1 + \alpha)$
P_8	(1)	$(1 + \alpha^2)$	(2α)
P_9	(1)	$(1 + \alpha + \alpha^2)$	$(1 + 2\alpha + \alpha^2)$
P_{10}	(1)	$(2 + \alpha^2)$	$(1 + 2\alpha + \alpha^2)$
P_{11}	(1)	(α)	$(2 + \alpha)$
P_{12}	(1)	$(1 + 2\alpha + \alpha^2)$	$(1 + \alpha)$
P_{13}	(1)	(1)	$(2 + 2\alpha^2)$
P_{14}	(1)	(0)	$(\alpha + \alpha^2)$

Table 1: A complete cap of $PG(2, 27)$ of size 14; each coordinate is an element of $\mathbb{F}_{27} = \mathbb{F}_3[\alpha]$, $\alpha^3 = \alpha + 1$, written as $c_0 + c_1\alpha + c_2\alpha^2$.

4 Machine verification

Two independent certificates are included in this submission:

- `reproduce.py`: a deterministic, stdlib-only Python script that rebuilds $\mathbb{F}_{27}$ and the 757 canonical points of $PG(2, 27)$, then verifies that the 14 listed points are canonical, distinct, pairwise non-collinear in triples, and that every outside point lies on a secant. It prints **PASS**.
- `lean4/Main.lean`: a Lean 4 (v4.33.1, no Mathlib, no `native_decide`) certificate in which

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theorem plane_size      : plane.length = 757                := by decide
theorem cap_small       : cap.length = 14 /\ cap.length < 21 := by decide
theorem cap_is_cap      : noThreeCollinear = true           := by decide
theorem cap_is_complete : isComplete = true                 := by decide
theorem main_disproof   : disproof_claim = true              := by decide

```

are proved purely by kernel reduction. `Check.lean` runs `#print axioms` on all five theorems; each reports “does not depend on any axioms” (zero axioms, zero `sorry`).